DHRUV RAJPUT

dhruvrajput3177@gmail.com | dhruv-rajput2007 | github.com/DHRUVRAJPUTTT | Patiala, India

EXPERIENCE

Reffro.com Freelance

Cloud Engineer & DevOps Remote

- Owned end-to-end production deployment on **AWS** using **EC2**, **S3**, and a custom **VPC** with properly segmented networking and access controls.
- Configured an **Application Load Balancer** with **Auto Scaling Groups**, scaling the platform to handle **10K+ concurrent active users** on **t3.small** instances.
- Automated data protection with scheduled **cron-based backup jobs** to **S3**, ensuring reliable recovery of application data.
- Right-sized compute and tuned scaling policies, reducing infrastructure costs by ~**20%**.

PROJECTS

Text-to-SQL Query Engine | *Python, FastAPI, React, OpenAI API, ChromaDB, LangChain*

- Built an end-to-end **RAG pipeline** embedding database schemas into a **ChromaDB** vector store and retrieving only the **top-3 relevant tables** per query, eliminating context-overflow issues and generating SQL in **under 2s** on average.
- Engineered a **semantic layer** annotating every column with business-context descriptions to guide the LLM toward accurate joins, aggregations, and filters.
- Implemented a **HITL safety guard** blocking non-SELECT SQL (INSERT, UPDATE, DELETE, DROP) and requiring explicit user confirmation before execution.
- Built a **React + TypeScript** frontend with syntax-highlighted SQL display, a pageable results table, and a schema explorer, connected to a **FastAPI** backend with observability logging.

Enterprise Agentic RAG Pipeline | *Python, LangGraph, FastAPI, Qdrant, Portkey*

- Built a production-grade **agentic RAG system** with **LangGraph** for cyclic reasoning, multi-step planning, and conversation memory, distinguishing true technical data from noisy inputs via semantic re-ranking.
- Implemented **NeMo Guardrails** to block off-topic, jailbreak, and injection inputs before retrieval, and routed all LLM calls through a **Portkey gateway** with automatic fallback between primary and backup Groq keys.
- Built an enterprise search layer using **Qdrant Cloud** vector search, **3072-dim Gemini embeddings**, and **FlashRank** local reranking, achieving ~**150ms** retrieval latency and **<3s** end-to-end response time.
- Developed a **RAGAS**-powered evaluation suite (**6 metrics**) with full observability via **Pydantic Logfire** and **LangSmith** tracing across every agent node.

Network Traffic Classification | *Python, Cybersecurity, ML*

- Designed and trained a machine learning model to classify network traffic for **cybersecurity threat detection**, analyzing **packet-level** and **flow-level** features to separate normal and malicious traffic.
- Applied supervised learning algorithms and feature engineering, bridging Computer Networks and ML concepts to address real-world security challenges.

TECHNICAL SKILLS

Languages: Java, C++, C, Python, Go (Golang)
AI / ML: LangChain, LangGraph, RAG, OpenAI API, ChromaDB, Qdrant, Prompt Engineering
Web & APIs: FastAPI, React, TypeScript, REST APIs
Cloud (AWS): EC2, S3, VPC, IAM, Load Balancers, Auto Scaling Groups
DevOps & Tools: Docker, CI/CD Pipelines, Cron, Linux, Git, GitHub
Core CS: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems

CERTIFICATIONS

- **AWS Agentic AI Demonstrated** – AWS Training & Certification (Jul 2026)
- **AWS Cloud Quest: Cloud Practitioner** – AWS Training & Certification (Aug 2026)
- **AWS Academy Graduate – Cloud Foundations** – AWS Academy, 20-hour program (Aug 2026)

EDUCATION

Thapar Institute of Engineering and Technology Patiala, India

B.Tech in Computer Science Aug. 2024 – June 2028